

Apache Commons StringUtils - Complete Reference

Package Import:

```
import org.apache.commons.lang3.StringUtils;
```

Maven Dependency:

```
<dependency>  
  <groupId>org.apache.commons</groupId>  
  <artifactId>commons-lang3</artifactId>  
  <version>3.14.0</version>  
</dependency>
```

Checking Methods

Method	Description
isEmpty()	Checks null or empty
isBlank()	Checks null, empty, whitespace
isNotBlank()	Opposite of isBlank
contains()	Checks substring
equals()	Null-safe equals
equalsIgnoreCase()	Ignore case compare
startsWith()	Checks prefix
endsWith()	Checks suffix
isNumeric()	Only numbers
isAlpha()	Only alphabets

Trim / Remove Methods

Method	Description
trim()	Remove leading/trailing spaces
strip()	Remove whitespace
deleteWhitespace()	Remove all whitespace
remove()	Remove substring
removeStart()	Remove from start
removeEnd()	Remove from end
chomp()	Remove line ending
chop()	Remove last character

Case Conversion Methods

Method	Description
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upperCase()	Convert uppercase
lowerCase()	Convert lowercase
capitalize()	First character uppercase
uncapitalize()	First character lowercase
swapCase()	Swap upper/lower case

Replace Methods

Method	Description
replace()	Replace substring
replaceOnce()	Replace first occurrence
replaceIgnoreCase()	Ignore case replace
replaceChars()	Replace characters
replaceEach()	Replace multiple strings

Split / Join Methods

Method	Description
split()	Split string
splitByWholeSeparator()	Split by separator
splitPreserveAllTokens()	Keep empty tokens
join()	Join list/array
joinWith()	Join using separator

Substring Methods

Method	Description
substring()	Extract substring
substringBefore()	Get before separator
substringAfter()	Get after separator
substringBetween()	Get between values
left()	Left chars
right()	Right chars
mid()	Middle chars

Padding / Repeat Methods

Method	Description
leftPad()	Add left padding
rightPad()	Add right padding
center()	Center align
repeat()	Repeat string

Reverse / Rotate Methods

Method	Description
reverse()	Reverse string
reverseDelimited()	Reverse tokens
rotate()	Rotate string

Difference / Similarity Methods

Method	Description
difference()	Difference between strings
indexOfDifference()	First different index
getCommonPrefix()	Find common prefix
getLevenshteinDistance()	Edit distance

Append / Prepend Methods

Method	Description
appendIfMissing()	Append suffix
prependIfMissing()	Add prefix

Counting / Search Methods

Method	Description
countMatches()	Count occurrences
indexOf()	Find index
lastIndexOf()	Last index
ordinalIndexOf()	Nth occurrence

Real Project Usage Examples

1. Validation: `if(StringUtils.isBlank(username)) { throw new RuntimeException("Username required"); }`
2. Safe Comparison: `StringUtils.equals(role, "ADMIN")`
3. CSV Handling: `StringUtils.join(list, ",")`
4. Formatting: `StringUtils.leftPad("7", 3, '0')`

Important Interview Difference

isEmpty()

Returns false for whitespace.

isBlank()

Returns true for whitespace-only strings.

Interview Tip: `StringUtils` provides null-safe utility methods for validation, parsing, formatting, comparison, and string transformation.